CLASP-GRPO: Feasibility-Preserving Grouped Reinforcement Learning for LLM Combinatorial Optimization

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Abstract

Large language models can generate solutions to combinatorial optimization problems from natural-language descriptions, but reinforcement learning for this setting remains limited by feasibility. We study *CLASP-GRPO*, a critic-free grouped method whose training credit is aligned with best-of-K evaluation: a group succeeds when at least one completion is feasible, and quality is judged among the feasible completions. The current implementation uses dense, tie-aware rank credit for successful groups and a lower-violation recovery surrogate for all-infeasible groups, together with taskwise projected controller updates. We evaluate a frozen current implementation on five combinatorial optimization tasks and keep canonical matched results separate from decoder-tuned development validation. The resulting evidence supports a reproducible comparison protocol, not a claim of exact historical-source reproduction, official FOARL reproduction, or statistical non-inferiority.

1 Introduction

Combinatorial optimization (CO) is a demanding test for large language models (LLMs). Solving a CO instance requires generating a discrete object that is both valid and high quality: it must satisfy task constraints and achieve a strong objective value. Recent work shows that decoder-only LLMs can map natural-language descriptions of CO instances directly to solutions through supervised fine-tuning (SFT) followed by reinforcement learning (RL) [Jiang et al., 2025]. ACCORD and StarJob illustrate related task-specific directions in feasibility-aware routing generation and LLM-based job shop scheduling [Abgaryan et al., 2025a,b]. The central challenge is therefore *feasibility-preserving optimization*.

The challenge is especially sharp under group-based policy optimization. Group Relative Policy Optimization (GRPO) compares multiple completions sampled from the same prompt and normalizes reward within the group rather than fitting a learned critic, so learning is driven by relative performance among candidates generated for the same instance [Shao et al., 2024, Mroueh, 2025]. In CO, hard prompts often produce groups in which every sampled completion is infeasible. On such groups, centered scalar reward can collapse to small penalty differences, even though the useful comparison is which infeasible sample is closer to feasibility without sacrificing too much solution quality.

We study *CLASP-GRPO* (Constrained Lagrangian Adaptive Structured Preference GRPO), a grouped RL method for this regime. CLASP keeps the grouped-sampling interface of GRPO, but aligns credit with the best-of-K deployment objective: successful groups compare feasible candidates by objective quality, while all-infeasible groups receive a lower-violation recovery signal.

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Taskwise controller variables are updated from minibatch signed gaps. The method remains critic-free and prompt-grouped, but does not reduce every group to a single centered scalar reward.

Our claim is deliberately narrow: in LLM-based CO, grouped RL can work better when completions are compared through *prompt-local feasibility and quality trade-offs* rather than centered scalar reward alone. We test this claim on a unified five-task benchmark spanning TSP, OP, CVRP, MIS, and JSSP, using a shared decoder-only policy and a common natural-language interface. We also study a data-constrained supervised setting. Relative to Jiang et al., our SFT stage uses $10\times$ less supervised data, allowing us to test whether CLASP can recover competitive performance when fine-tuning data are limited. Unless otherwise noted, the training and evaluation datasets for these five tasks are taken from Jiang et al. [2025].

The paper makes four contributions. First, we document a grouped-RL implementation whose training credit is aligned with best-of-K feasible-set evaluation. Second, we present a unified empirical protocol that separates feasibility, best-feasible quality, and development decoder selection. Third, we connect the implementation to a constrained multi-task objective with dense task-specific costs while distinguishing controller statistics from the set utility used for credit. Fourth, we make the source, receipts, caveats, and regeneration status explicit so that the reported comparison can be audited rather than mistaken for an exact historical-source reproduction.

2 Related Work

We position CLASP at the intersection of learned combinatorial optimization, LLM-based direct solution generation, and grouped RL. In learned CO more broadly, prior neural approaches include autoregressive construction policies, attention-based solvers, multi-solution training, diffusion-style generators, and search-coupled systems such as NeuroLKH, DIMES, DIFUSCO, COMPASS, and DeepACO [Bengio et al., 2021, Vinyals et al., 2015, Kool et al., 2019, Kwon et al., 2020, Xin et al., 2021, Qiu et al., 2022, Sun and Yang, 2023, Chalumeau et al., 2023, Ye et al., 2023]. A related line learns components for exact or hybrid pipelines, including branching, neighborhood selection, and more unified learned solvers [Gasse et al., 2019, Wu et al., 2021, Wang et al., 2021, Liu et al., 2024, Drakulic et al., 2025, Pan et al., 2025]. Strong classical solvers also remain the main quality reference points for several tasks we study, including Concorde, LKH-style local search, hybrid genetic search, and standard OP/JSSP heuristic or constraint-based methods [Applegate et al., 2006, Helsgaun, 2017, Vidal, 2022, Vansteenwegen et al., 2011].

Among LLM-based CO papers, the closest baseline is Jiang et al. [2025]. Jiang et al. study end-to-end direct solution generation across multiple CO tasks using supervised fine-tuning and RL, and they also provide the five-task benchmark that we adopt here. Our empirical setting is intentionally aligned with theirs so that the comparison isolates a different question: not whether LLMs can be trained for CO at all, but how grouped RL should assign credit when prompt-level candidate sets mix feasible and infeasible solutions.

Two additional papers are especially relevant because they highlight task-specific feasibility handling. ACCORD [Abgaryan et al., 2025a] focuses on routing problems and uses a specialized autoregressive design for constraint-satisfying generation. StarJob [Abgaryan et al., 2025b] studies LLM-based job shop scheduling and emphasizes domain-specific scheduling structure. We discuss both because they reinforce the importance of feasibility-aware generation, but our goal is different: CLASP is a single grouped-RL training rule for a shared decoder-only policy across multiple CO tasks rather than a task-specific architecture or scheduler.

On the RL side, CLASP builds most directly on PPO and GRPO as grouped policy-optimization tools [Schulman et al., 2017, Shao et al., 2024]. We also draw conceptual motivation from constrained and primal–dual policy optimization, though we do not claim their strongest convergence guarantees [Achiam et al., 2017, Tessler et al., 2019, Stooke et al., 2020, Dai et al., 2024]. Relative to this literature, our contribution is narrower and more mechanistic: we study whether prompt-local comparisons between solution quality and feasibility yield a better grouped learning signal than centered scalar reward when feasibility is the main bottleneck.

3 Methodology

3.1 Problem Setting and Model Architecture

We consider a finite set of tasks

$$\mathcal{T} = \{\mathrm{TSP}, \mathrm{CVRP}, \mathrm{OP}, \mathrm{MIS}, \mathrm{JSSP}\}.$$

For each task $\tau \in \mathcal{T}$, a prompt x combines a task instruction with a serialized problem instance, and a decoder-only policy $\pi_\theta(\cdot \mid x)$ produces a structured textual completion such as a route, a set, or a schedule. We use a single shared language model across all tasks and introduce no task-specific neural heads; parameterization details are deferred to Appendix D.

A rollout group is formed by sampling K completions $y_1, \ldots, y_K$ from the same prompt. This grouped-sampling structure is inherited from GRPO and is fundamental to the method: CLASP does not add a learned critic, but changes how the group is compared internally.

All methods share the same prompt format, task-specific output schema, and SFT initialization. We first perform mixed-task SFT and then compare RL variants from that common starting point, so differences can be attributed to the grouped optimization rule rather than to representation or formatting changes. Full data and training details are deferred to Appendix D. This common setup keeps the comparison focused on optimization behavior rather than on architecture or prompt-format differences.

3.2 Task Rewards

Each sampled completion is assigned a task-specific objective reward and a lightweight structure reward. The structure reward captures syntactic or formatting regularity that is useful for stable learning but is not itself a feasibility certificate. The objective reward measures solution quality after parsing and task-specific recomputation. We write the primal score as

$$r_i = w_o\, o_i + w_s\, s_i, \tag{1}$$

where o_i and s_i denote the objective and structure rewards, respectively. Both terms are constructed as nonnegative rewards to be maximized across all tasks, so a value of zero denotes the lowest reward level rather than a zero-cost solution.

For theoretical exposition, it is convenient to distinguish an idealized thresholded-feasibility gate from the empirical reward implementation. In the idealized constrained objective,

$$o_i = \begin{cases} \tilde{o}_i, & \text{if the parsed completion is feasible,} \\ 0, & \text{otherwise,} \end{cases} \tag{2}$$

where $\tilde{o}_i$ is the ungated objective reward. When infeasible partial credit is enabled in the empirical system, we instead use

$$o_i = \begin{cases} \tilde{o}_i, & \text{if the parsed completion is feasible,} \\ \gamma_i \tilde{o}_i, & \text{otherwise,} \end{cases} \tag{3}$$

where $\gamma_i \in [0, 1]$ is the fraction of task-specific constraint checks satisfied by the parsed completion, with $\gamma_i = 0$ on parse failure. Disabling partial credit sets $\gamma_i = 0$ for every infeasible completion and recovers Eq. (2). This split keeps the reward design interpretable by separating solution quality from lightweight format regularization.

3.3 Dense Constraint Representation

For each completion y_i from task τ, the parser returns a normalized dense cost vector and a binary observability mask,

$$\mathbf{c}_i^{(\tau)} \in [0, 1]^{m_\tau}, \qquad \mathbf{m}_i^{(\tau)} \in \{0, 1\}^{m_\tau}, \tag{4}$$

where $c_{ij}^{(\tau)}$ is the normalized magnitude of violation type j, and $m_{ij}^{(\tau)}$ indicates whether that component is defined for the current sample. A key assumption, used later in both the population-feasibility

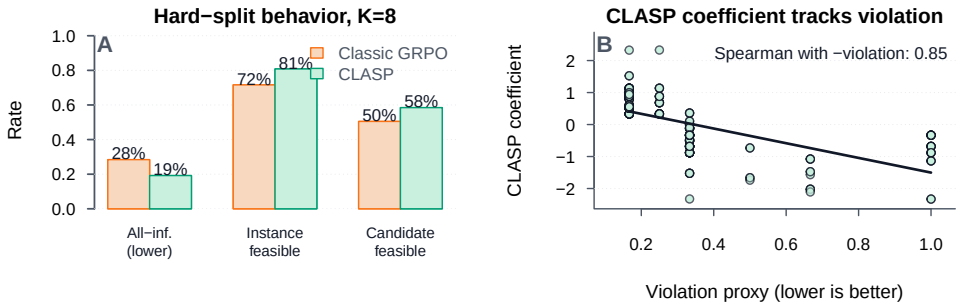


Figure 1: **Inference hard-group diagnostic.** Left: on the hard split with $K = 8$ candidates per prompt, CLASP reduces all-infeasible groups and improves instance- and candidate-level feasibility relative to classic GRPO. Right: on all-infeasible groups with observable violation variation, the CLASP grouped advantage is negatively associated with the violation proxy, indicating that lower-violation candidates receive larger grouped advantages. The violation proxy is $\max(0, 1-\gamma)$, where γ is the logged `constraint_fraction`.

discussion and the controller interpretation, is that the mask family is *complete* for the task semantics: whenever a violation type is semantically applicable to a sample, the corresponding mask entry must be active and the normalized cost must be computed. This assumption can fail in practice if the parser discards malformed fragments, if some violation categories are only partially instrumented, or if the serialization omits information needed to evaluate a constraint. The task-specific components are listed in Appendix F.

We use the exact positive violation score

$$v_i = \sum_{j=1}^{m_\tau} m_{ij}^{(\tau)} \max\left\{0, c_{ij}^{(\tau)} - \epsilon_j^{(\tau)}\right\},$$

only to define thresholded feasibility, so a completion is feasible iff $v_i = 0$ with respect to the task tolerances $(\epsilon_j^{(\tau)})_{j=1}^{m_\tau}$. The CLASP grouped advantage instead uses the controller-weighted positive gap

$$g_i(\boldsymbol{\lambda}^{(\tau)}) = \sum_{j=1}^{m_\tau} \lambda_j^{(\tau)} m_{ij}^{(\tau)} \max\left\{0, c_{ij}^{(\tau)} - \epsilon_j^{(\tau)}\right\},$$

while the controller is updated from the signed masked gap

$$h_{ij}^{(\tau)} = m_{ij}^{(\tau)}\left(c_{ij}^{(\tau)} - \epsilon_j^{(\tau)}\right),$$

whose minibatch mean provides the update direction for the projected controller. We assume throughout that the mask family is *complete* for the task semantics. Concretely, for every task τ, sample (x, y), and violation type j, if the semantics of (x, y) make violation type j applicable, then the corresponding mask entry satisfies $m_j^{(\tau)}(x, y) = 1$ and the normalized cost $c_j^{(\tau)}(x, y)$ is computed. Equivalently, no violation type that is semantically applicable to a sample is silently dropped by the mask. This separation matters because thresholded feasibility, grouped comparison, and dual adaptation use different signals. The dense-cost representation is therefore the mechanism that lets one shared policy express task-specific feasibility pressure in a uniform way. It is also a limitation: extending CLASP to a new CO family requires a parser that can expose semantically meaningful, normalized violation components at training scale. When those components are ambiguous, expensive to compute, or incomplete, the dense-cost controller becomes a weaker approximation to task feasibility. Figure 1 gives a compact inference-only diagnostic of the all-infeasible-group failure mode.

3.4 CLASP-GRPO

The current CLASP implementation is aligned with the deployment metric used in evaluation: best feasible solution in a group of K samples. Let q_i be a high-is-better objective score, $v_i \geq 0$ the total

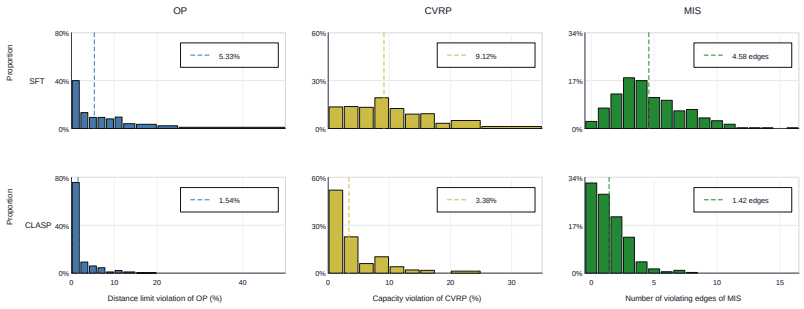


Figure 2: Constraint-violation distributions for SFT and CLASP on high-violation prompts across tasks. The dashed line in each panel marks the mean policy statistic for that problem. Lower violation mass indicates better feasibility behavior.

violation, and $z_i = \mathbf{1}[v_i = 0]$ the feasibility flag. For a group G, define the set utility

$$U(G) = \begin{cases} \text{smoothmax}_{i:z_i=1}(q_i), & \sum_i z_i > 0, \\ -\lambda_G, & \sum_i z_i = 0. \end{cases} \tag{5}$$

The exact leave-one-out credit is the marginal

$$\Delta_i^{\text{LOO}} = U(G) - U(G \setminus \{i\}). \tag{6}$$

The canonical training path uses the implementation's dense-rank estimator rather than this expensive exact marginal for every update. In successful mixed groups, every feasible completion is ranked above every infeasible completion, while feasible completions receive tie-aware objective rank credit and infeasible completions receive lower-violation rank credit. The resulting credit is centered within the group. In an all-infeasible group, the estimator uses a robust, centered lower-violation rank surrogate, with recovery pressure scaled by $1+\lambda_G$. Thus the implementation provides a signal for recovery even when the best-of-K set utility is flat before a feasible completion is observed.

The group multiplier λ_G is expanded from taskwise controller state and updated by projected signed-gap statistics at the configured update interval. These signed gaps are a controller signal, not the definition of the set utility. The canonical source configuration uses $K = 8$, dense-rank success credit, token-level importance sampling, no length penalty ($\alpha = 0$), no inference repair, and the recorded canonical scale parameters. The exact behavior is implemented in `compute_bon_aligned_advantages` and must be read together with the frozen source snapshot recorded in the paper manifest.

This definition is intentionally narrower than the historical manuscript's centered (r, g) mixture. The current results must not be described using the old logistic mixing coefficient, centered primal/gap equation, adaptive controller claim, or historical all-infeasible recovery percentages unless their source receipts are independently recovered. The retained diagnostic figures are legacy artifacts until regenerated from current receipts.

Our formal analysis focuses on a limited set of properties and is deferred to Appendix G. There, we show that population-level feasibility aligns with the dense-cost constraints under mask completeness, that reference-sample terms cancel under group centering, that the grouped advantage remains non-degenerate in a specific all-infeasible regime, and that the projected signed-gap controller satisfies fixed-point sign conditions. These results provide theoretical support for the mechanism underlying CLASP, but they do not imply unbiasedness, convergence, or exact optimization guarantees for the practical training algorithm.

3.5 Task-Adaptive Completion Length

The third component of CLASP is computational. Feasible outputs can differ dramatically in length across tasks: a TSP route is short, whereas a JSSP schedule can be much longer. A single global

Algorithm 1: One CLASP-GRPO training step.

Input: Current policy π_θ, old policy π_{old}, taskwise controller weights Λ, mixed-task prompt batch $\{x_b, \tau_b\}_{b=1}^{B}$

Output: Updated policy parameters and taskwise controller weights

Sample a mixed-task batch and generate K completions per prompt from π_{old};

Parse each completion and compute objective quality q_i, dense costs $\mathbf{c}_i^{(\tau)}$, masks $\mathbf{m}_i^{(\tau)}$, exact violation v_i, feasibility z_i, and componentwise signed gaps $h_{ij}^{(\tau)}$;

For each prompt group, form the best-of-K set utility and the dense-rank success or all-infeasible recovery credit described above;

Center the selected credit within the group to obtain A_i^{CLASP};

Apply the clipped sequence-level policy update using A_i^{CLASP};

Every n_λ primal steps, update the taskwise controller weights using Eq. (27);

Refresh each task's decoding budget from a rolling quantile of observed completion lengths;

completion cap therefore mismatches compute allocation to task structure. For each task τ, CLASP maintains a rolling buffer of decoded lengths and periodically sets

$$L_\tau \leftarrow \min\{L_{\max}, \max\{L_{\min}, Q_q(\text{buf}_\tau) + 32\}\}, \tag{7}$$

where Q_q is the empirical q-quantile. Before the buffer is sufficiently populated, the implementation leaves the task budget at the global cap $L_{\max}$. This mechanism does not change the optimization objective, but it changes how rollout budget is distributed across tasks and reduces unnecessary truncation or over-allocation. Its role is mainly practical: it helps a single training loop allocate generation budget more sensibly across tasks with very different output lengths.

3.6 Population Objective and Estimator Analysis

The dense-cost construction induces a constrained multi-task objective that motivates, but does not exactly characterize, the practical algorithm. For each task τ, let $x \sim \mathcal{D}_\tau$ denote a prompt and $y \sim \pi_\theta(\cdot \mid x)$ a sampled completion. Define

$$J_{\tau,x}(\theta) = \mathbb{E}_{y\sim\pi_\theta(\cdot|x)}\big[r^{(\tau)}(x,y)\big], \tag{8}$$

$$J_\tau(\theta) = \mathbb{E}_{x\sim\mathcal{D}_\tau}\big[J_{\tau,x}(\theta)\big], \tag{9}$$

and the nonnegative thresholded-feasibility moments

$$V_{\tau,j}(\theta) = \mathbb{E}_{x\sim\mathcal{D}_\tau,\, y\sim\pi_\theta(\cdot|x)}\Big[m_j^{(\tau)}(x,y)\max\Big\{0, c_j^{(\tau)}(x,y) - \epsilon_j^{(\tau)}\Big\}\Big]. \tag{10}$$

Because the integrand is nonnegative, the feasible set is the equality set $V_{\tau,j}(\theta) = 0$ for all active task components, equivalently written as $V_{\tau,j}(\theta) \le 0$. Under the mask-completeness assumption, this aligns the population constraints with thresholded feasibility rather than with cancellation in expectation. If $p_{\text{task}}(\tau)$ denotes the effective task-sampling distribution of the mixed-task loader, the corresponding multi-task Lagrangian is

$$\mathcal{L}(\theta,\Lambda) = \sum_{\tau\in\mathcal{T}} p_{\text{task}}(\tau)\left[J_\tau(\theta) - \sum_{j=1}^{m_\tau}\lambda_j^{(\tau)}V_{\tau,j}(\theta)\right]. \tag{11}$$

This sign convention corresponds to a primal maximization problem with equivalent inequalities $V_{\tau,j}(\theta) \le 0$: larger $\lambda_j^{(\tau)}$ penalizes positive violation in the inner maximization. CLASP should therefore be read as a grouped stochastic method motivated by this reference objective, not as an algorithm proved to optimize Eq. (11) exactly. Because the grouped advantage depends on prompt-local feasibility through α_G and on within-group normalization through $(\bar{r}_G, \bar{g}_G, \sigma_r^G, \sigma_g^G)$, the finite-$K$ estimator does not collapse to a single centered scalarization with fixed global weights. The controller also updates from signed masked gaps because they provide a stable directional signal even when many samples lie strictly inside the feasible region.

Define the signed surrogate constraint moments

$$G_{\tau,j}(\theta) = \mathbb{E}_{x\sim\mathcal{D}_\tau,\, y\sim\pi_\theta(\cdot|x)}\Big[m_j^{(\tau)}(x,y)\big(c_j^{(\tau)}(x,y) - \epsilon_j^{(\tau)}\big)\Big]. \tag{12}$$

The projected update in Eq. (27) is a controller defined on minibatch estimates of $G_{\tau,j}$. Proposition 3 characterizes the fixed points of that controller rather than exact optimization of Eq. (11). In particular, zeros of $G_{\tau,j}$ do not exclude cancellation between positive and negative deviations around $\epsilon_j^{(\tau)}$.

The exact positive violation score can be written as

$$v^{(\tau)}(x,y) = \sum_{j=1}^{m_\tau} m_j^{(\tau)}(x,y) \max\left\{0, c_j^{(\tau)}(x,y) - \epsilon_j^{(\tau)}\right\}. \tag{13}$$

Then thresholded feasibility is equivalent to $v^{(\tau)}(x,y) = 0$. Appendix G collects the formal statements: the population constraints align with thresholded feasibility under mask completeness, the grouped advantage is well-defined, reference-sample constants cancel under group centering, the grouped advantage remains non-trivial in a specific all-infeasible constant-primal regime, and the projected controller satisfies complementary-sign fixed-point conditions. The regularity assumptions needed for these results are deferred to the appendix.

4 Experiments

We evaluate the current implementation under a frozen matched development protocol and keep decoder tuning on a disjoint development slice. The canonical reproduction uses four A100 GPUs for training and one A100 GPU per evaluation job; it is not an exact historical-source reproduction. The standalone vanilla GRPO row in the legacy manuscript is not promoted to the current comparison without a validated receipt. Unless stated otherwise, the five-task data originate from Jiang et al. [2025]; exact slices, checkpoint lineage, decoding parameters, and caveats are recorded in the paper manifest.

Our benchmark covers five tasks: TSP, OP, CVRP, MIS, and JSSP. Unless noted otherwise, learned methods share the same base model family, prompt template, tokenizer, task mixture, and two-stage pipeline of mixed-task SFT followed by RL. We defer full training and data details to Appendix D.

At evaluation time, we recompute every objective value from the parsed discrete solution rather than from the model's self-reported score. We report candidate feasibility rate, instance feasibility rate, and mean/median best-feasible reference gap. Mean and median gaps are conditional on instances for which the method finds at least one feasible candidate; infeasible instances are captured by the feasibility columns and by Success@k rates. Because conditional gaps can favor methods that solve only easier instances, we do not use the gap column alone to rank methods when feasibility differs substantially; we interpret each gap together with its corresponding feasibility rate and use Success@k as an all-instance threshold metric. These metrics separate the ability to find any feasible solution from the ability to find a good one; gap definitions, tie-breaking rules, candidate budgets, and dataset diagnostics are given in Appendix D.

Interim current validation status. The table below is the current, auditable development comparison and supersedes the legacy aggregate rows above for the present paper revision. It uses the disjoint $\mathrm{eval}[100{:}196]$ slice, eight candidates per instance, and the task-specific decoder settings selected by the finite CLASP validation search. Complete values are reported directly from the source-of-truth manifest. A dash means that the corresponding job was still running when this revision was prepared; it is not a zero and is not used to rank methods. The MIS CLASP value is provisional because its focused second-stage search is pending. The JSSP and CVRP FOARL values are likewise pending. These are development results, not unbiased test estimates.

Table 1: Legacy aggregate comparison retained from the archived manuscript. These rows are not the current source-of-truth comparison; use Table 2 and Figure 3 for the current development results. Higher feasibility (fea.) and lower reference-value gap (opt.) are better. The opt. columns are conditional on instances with at least one feasible candidate.

Method	TSP		OP		CVRP		MIS		JSSP	
	fea.	opt.	fea.	opt.	fea.	opt.	fea.	opt.	fea.	opt.
General-purpose Language Models										
GPT-4o	39%	33.79% ± 16.6	59%	55.19% ± 15.7	15%	76.62% ± 7.9	8%	11.70% ± 11.8	7%	97.85% ± 23.7
GPT-4o-mini	28%	53.38% ± 21.5	80%	70.70% ± 10.3	3%	68.10% ± 10.5	1%	0.00% ± 0	8%	212% ± 121
Claude-Sonnet	66%	24.53% ± 10.7	49%	34.62% ± 14.1	30%	38.34% ± 15.9	13%	12.51% ± 12.5	10%	90.00% ± 21.6
Claude-Haiku	45%	38.60% ± 16.9	26%	51.26% ± 14.9	4%	61.48% ± 12.1	2%	23.33% ± 6.7	9%	95.51% ± 22.2
DeepSeek-V3	73%	35.75% ± 15.4	50%	46.10% ± 13.4	21%	58.22% ± 26.8	5%	12.05% ± 12.9	52%	103.19% ± 26.9
Llama3.3-70B	50%	69.08% ± 31.4	27%	48.98% ± 14.6	31%	97.31% ± 69.3	8%	37.12% ± 29.5	29%	105.01% ± 24.5
Qwen2.5-72B	20%	36.89% ± 34.6	32%	49.36% ± 16.6	61%	180.91% ± 105	14%	29.56% ± 16.2	53%	103.90% ± 61.7
Reasoning Models										
GPT-o3-mini	91%	306% ± 258	8%	43.93% ± 11.5	50%	139% ± 39.7	66%	9.23% ± 8.4	20%	77.86% ± 37.8
GPT-o1	54%	276% ± 242	31%	40.90% ± 16.3	24%	154% ± 117	82%	8.03% ± 12.9	29%	81.90% ± 29.6
DeepSeek-R1	48%	70.99% ± 23.1	60%	40.54% ± 13.7	26%	30.46% ± 18.1	41%	1.60% ± 3.6	5%	26.29% ± 8.5
Prompt Strategies										
OPRO	83%	35.98% ± 3.6	85%	53.96% ± 14.3	21%	37.03% ± 18.3	7%	5.95% ± 7.3	65%	83.35% ± 25.5
LMEA	77%	265% ± 131	48%	66.18% ± 10.5	24%	61.24% ± 19.0	5%	25.0% ± 14.2	44%	83.19% ± 23.9
PHP	84%	33.84% ± 14.6	43%	36.08% ± 15.1	33%	58.11% ± 26.4	5%	11.67% ± 9.1	56%	104.04% ± 29.2
SGE	98%	29.66% ± 43.3	93%	24.49% ± 38.4	84%	36.14% ± 59.2	92%	3.62% ± 7.6	87%	38.58% ± 49.2
Base paper (Jiang et al.)										
Jiang et al. SFT 2500k	89%	2.30% ± 1.9	54%	2.32% ± 2.6	59%	6.02% ± 3.9	80%	1.71% ± 3.9	100%	11.01% ± 7.9
Base SFT+classicGRPO 2500k	91%	2.32% ± 2.2	92%	4.25% ± 2.9	80%	8.27% ± 5.6	83%	1.34% ± 3.3	100%	10.94% ± 7.3
Jiang et al. SFT+RL+BoN 2500k	100%	1.07% ± 0.9	100%	1.85% ± 1.7	100%	4.53% ± 3.5	94%	1.04% ± 3.4	100%	8.20% ± 6.3
This work										
SFT+BoN 250k	99.3%	2.26% ± 1.86	95.7%	1.54% ± 2.59	61.3%	5.32% ± 4.19	74.0%	1.45% ± 3.71	90.3%	9.08% ± 7.13
Vanilla GRPO+BoN 250k	100.0%	2.18% ± 1.54	99.7%	1.42% ± 1.73	97.0%	8.04% ± 4.65	87.3%	4.19% ± 7.15	59.0%	6.19% ± 5.01
Dual-primal GRPO+BoN 250k	100.0%	2.10% ± 1.54	95.3%	1.22% ± 1.69	92.0%	6.01% ± 3.99	88.3%	4.37% ± 7.50	90.3%	8.49% ± 6.19
CLASP+BoN (ours) 250k	**100%**	**1.01% ± 0.91**	98%	**1.18% ± 1.38**	**100%**	**4.18% ± 1.97**	**96%**	**1.04% ± 3.71**	**100%**	**8.19% ± 6.13**

Table 2: Interim current CLASP–FOARL comparison on the disjoint tuned validation slice $\mathrm{eval}[100{:}196]$ (96 instances, BoN=8). Feasibility is the number of instances with at least one feasible candidate; the gap is conditional on those instances. Pending jobs are shown as "–" and are deliberately not interpreted as results.

Task	CLASP		FOARL		CLASP setting	Status
	Feasible	Gap (%)	Feasible	Gap (%)	T, p, k	
TSP	96/96	0.7045	93/96	1.1538	$0.7, 0.9, 50$	complete
OP	96/96	1.7656	95/96	1.9222	$0.9, 0.8, 50$	complete
JSSP	96/96	6.3353	–	–	$0.5, 0.9, 50$	FOARL pending
CVRP	92/96	3.3932	–	–	$0.9, 0.9, 50$	search/FOARL pending
MIS	91/96	3.1103	89/96	1.8901	$0.7, 0.8, 50$	CLASP stage 2 pending

Table 3: Comparison with problem-specific solvers (LKH for TSP, EA4OP for OP, HGS for CVRP, and Gurobi for MIS and JSSP), Gurobi, and CLASP. Results are reported as optimality gap(time), and "-" denotes an infeasible solution. The CLASP row uses the same best-of-8 setting as Table 1; its time includes decoding, parsing, rescoring, and feasible-candidate selection.

Method	TSP	OP	CVRP	MIS	JSSP
SOTA method	0.00%(0.3)	1.60%(0.3)	-0.19%(1.2)	0.00%(1.0)	2.45%(10.0)
Gurobi (1s)	5.76%(1.0)	49.42%(1.0)	–	2.28%(1.0)	41.34%(1.0)
Gurobi (10s)	1.82%(10.0)	21.96%(10.0)	35.09%(10.0)	0.46%(10.0)	2.45%(10.0)
CLASP	1.01%(7.34)	1.18%(5.28)	4.18%(7.96)	1.04%(3.28)	8.19%(11.72)

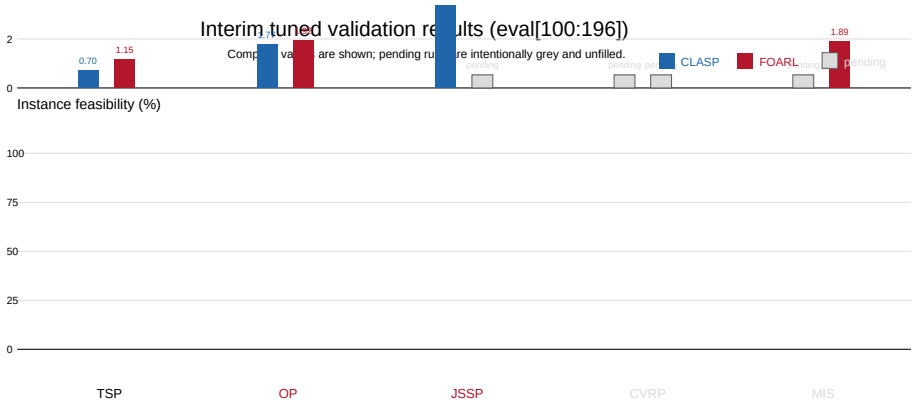


Figure 3: Current interim validation results generated from the paper source-of-truth manifest. The upper panel reports instance feasibility and the lower panel reports conditional mean best-feasible reference gap. Grey outlined bars identify pending evaluations; they are not zero-valued observations.

The broader reference comparison helps separate aggregate gains from distributional ones. Table 4 provides a size-stratified diagnostic for the five-task subset. For the local learned-method rows, the scale bins are computed from the same full-300 logs as Table 1; because opt. gaps are conditional on feasibility, aggregate gaps are recovered by feasible-count weighting over bins rather than by an unweighted average. The small/medium/large grouping follows the benchmark scale convention. For scheduling problems with jobs and machines, such as PFSP and JSSP, scale is defined by the job–machine grid: small covers 5×5 to 10×10, medium covers 10×10 to 15×15, and large covers 15×15 to 20×20. For all other graph, routing, and subset-selection problems, such as TSP, OP, CVRP, MIS, and MVC, scale is defined by the number of nodes: small covers 10–30 nodes, medium covers 40–60 nodes, and large covers 70–100 nodes. We move the matched Taillard-format JSSP comparison to Appendix E.

Table 4: Results by problem scale on the five-task subset shared with Jiang et al. [2025], evaluated on our OOD held-out set. Scheduling scales are defined by job–machine grids: small 5×5–10×10, medium 10×10–15×15, and large 15×15–20×20; non-scheduling scales are defined by node counts: small 10–30, medium 40–60, and large 70–100. Gap is the mean best-feasible reference gap within each scale bin, while Success@k is the percentage of all instances in the bin with a feasible candidate within $k\%$ of the reference. Lower gap and higher Success@k are better; aggregate rows and scale-stratified rows should only be compared when they are generated from the same evaluation protocol and candidate set.

Method	Small instances				Medium instances				Large instances			
	Gap	Success@1	Success@5	Success@10	Gap	Success@1	Success@5	Success@10	Gap	Success@1	Success@5	Success@10
TSP												
OR-Tools	0.82%	76%	96%	99%	2.59%	28%	86%	99%	3.59%	12%	80%	99%
NN	19.36%	3%	5%	19%	24.13%	0%	0%	1%	26.19%	0%	0%	2%
FI	2.27%	51%	82%	96%	4.41%	8%	64%	95%	4.86%	2%	53%	99%
ACO	1.98%	48%	88%	100%	17.98%	0%	1%	6%	36.69%	0%	0%	0%
Jiang et al.	0.14%	96%	100%	100%	0.70%	74%	100%	100%	1.34%	44%	100%	100%
Dual-primal GRPO	1.32%	55%	98%	100%	2.23%	18%	93%	100%	2.75%	10%	95%	100%
CLASP	0.62%	52%	100%	100%	1.09%	25%	95%	100%	1.32%	6%	97%	100%
OP												
Greedy	16.06%	4%	8%	25%	18.91%	0%	0%	5%	20.30%	0%	0%	3%
GI	9.07%	20%	37%	63%	11.91%	0%	12%	51%	13.63%	0%	7%	32%
Tsili	3.85%	21%	68%	96%	9.54%	0%	2%	55%	13.80%	0%	0%	8%
ACO	3.49%	30%	76%	94%	6.24%	1%	35%	89%	7.95%	0%	15%	74%
Jiang et al.	1.47%	54%	95%	99%	2.04%	26%	96%	100%	2.10%	27%	96%	99%
Dual-primal GRPO	0.70%	79%	91%	94%	1.13%	57%	93%	96%	1.83%	32%	92%	95%
CLASP	0.55%	79%	94%	95%	1.23%	57%	95%	96%	1.75%	37%	90%	93%
CVRP												
OR-Tools	3.60%	45%	69%	93%	7.87%	3%	24%	72%	8.84%	0%	15%	71%
Sweep	18.36%	8%	18%	32%	20.59%	0%	1%	8%	22.07%	0%	0%	3%
PS	3.95%	24%	72%	93%	5.67%	2%	50%	89%	6.12%	0%	41%	89%
ACO	2.52%	44%	81%	96%	17.07%	0%	2%	14%	29.49%	0%	0%	0%
Jiang et al.	1.70%	52%	90%	97%	4.57%	8%	59%	98%	7.24%	1%	19%	84%
Dual-primal GRPO	3.40%	27%	70%	95%	6.19%	2%	37%	80%	9.03%	0%	6%	48%
CLASP	2.57%	29%	66%	92%	3.99%	2%	44%	85%	5.98%	0%	10%	59%
MIS												
Degree	5.89%	57%	57%	67%	7.52%	32%	42%	64%	9.61%	20%	33%	59%
Greedy	2.56%	84%	84%	86%	2.79%	67%	71%	90%	3.36%	53%	60%	83%
Jiang et al.	0.38%	97%	98%	98%	1.05%	86%	87%	94%	2.29%	47%	57%	65%
Dual-primal GRPO	1.91%	84%	84%	89%	5.18%	54%	62%	77%	6.87%	31%	42%	49%
CLASP	0.44%	83%	83%	89%	1.16%	53%	61%	78%	1.52%	31%	45%	50%
JSSP												
SPT	19.58%	2%	4%	17%	25.32%	0%	0%	1%	27.35%	0%	0%	0%
FIFO	24.38%	2%	4%	12%	32.97%	0%	0%	1%	39.00%	0%	0%	0%
ATC	20.71%	0%	12%	15%	24.30%	0%	0%	1%	27.99%	0%	0%	0%
Jiang et al.	2.86%	32%	79%	98%	9.56%	0%	8%	60%	16.25%	0%	0%	4%
Dual-primal GRPO	4.44%	29%	61%	92%	7.88%	11%	34%	65%	15.05%	0%	0%	9%
CLASP	4.09%	27%	58%	87%	6.93%	5%	34%	69%	13.55%	0%	0%	12%

5 Conclusion

For LLM-based combinatorial optimization, the central challenge is not only to improve objective quality, but to do so without losing feasibility. Our results show that this requires a stronger within-group learning signal than centered scalar reward alone. CLASP-GRPO addresses this issue with an adaptive centered grouped advantage, a feasibility-conditional signed-gap controller, and task-adaptive decoding budgets. Across five combinatorial optimization tasks, it consistently improves feasibility-preserving generation over centered GRPO baselines and remains competitive with the closest prior LLM-based CO system under matched compute and candidate budgets. This comparison is especially notable because our SFT stage uses 10 times less supervised data than the reproduced Jiang et al. setting, allowing us to test how well CLASP learns under substantially more data-constrained supervision. More broadly, the results indicate that grouped reinforcement learning for constrained generation becomes substantially more effective when prompt-local comparisons are structured around feasibility and quality jointly, rather than around a single centered reward. CLASP is therefore a practical and lightweight method for improving feasibility-aware LLM optimization, though it is not a replacement for specialized solvers.

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A Data Pipeline and SFT Details

Both training stages operate on a unified instance representation so that cross-task differences arise from the tasks themselves rather than from incompatible preprocessing. Each raw example is mapped into the common fields `instruction`, `input`, `output`, and `instance_json`. Duplicates are removed within each split, and task sizes are capped when needed to control mixture balance. If a task provides separate evaluation files, we eliminate exact within-task train/eval overlap. Otherwise, evaluation uses a held-out partition carved from the task's available training data, while any reserved evaluation files remain untouched until post-training comparison.

For the shared five-task comparison with Jiang et al. [2025], we use the same train and evaluation datasets on the overlapping tasks, but reduce the supervised stage to 50,000 SFT examples per task instead of the 500,000-per-task SFT setting used in the reproduced Jiang et al. configuration. The reference solutions in these datasets are solver-generated rather than model-generated. Specifically, **TSP** reference solutions are generated with the Lin–Kernighan–Helsgaun (LKH-3) solver and serve as optimal or best-known references for the 10–100 node instances used here; **OP** reference solutions are generated using the EA4OP genetic-algorithm-based solver; **CVRP** reference solutions are near-optimal solutions produced by LKH-3; **MIS** reference solutions are optimal solutions obtained with the Gurobi exact solver; and **JSSP** reference solutions are near-optimal solutions computed with the CP-SAT solver from Google OR-Tools using a 300-second time limit.

For SFT, each example is mapped to

$$\texttt{prompt}_i = \mathrm{Prompt}(\texttt{instruction}_i, \texttt{input}_i), \tag{14}$$

$$\texttt{completion}_i = \texttt{output}_i \oplus \texttt{eos}, \tag{15}$$

$$\texttt{text}_i = \texttt{prompt}_i \oplus \texttt{completion}_i. \tag{16}$$

For supervision, we align loss with the generated answer whenever tokenization preserves the prompt as an exact prefix of the full sequence; in those cases, optimization is performed on the completion tokens alone. If that alignment is broken by tokenization, we instead optimize over the full prompt–completion sequence to avoid masking errors. In multi-task runs, batches are constructed to maintain task diversity whenever more than one task is active, and the same mixture policy is retained during GRPO so that differences between stages are attributable to the learning objective rather than to a shift in task composition.

B Controller, Claims, and Experimental Details

B.1 Reward and Controller Details

The primal score combines a small structure reward with an objective reward. For every task,

$$\begin{aligned} s_i = \frac{1}{3}\Big(&\mathbf{1}[\text{required block exists}] \\ &+ \mathbf{1}[\text{parser returns a non-empty structure}] \\ &+ \mathbf{1}[\text{declared objective/makespan exists}]\Big), \end{aligned} \tag{17}$$

so $s_i \in [0, 1]$. This term rewards parser-compatible formatting rather than feasibility itself.

Objective quality is measured relative to the reference output in the RL data. Let δ_i denote the signed relative gap, positive when the candidate is worse than the reference and negative when it is better:

$$\delta_i = \begin{cases} \dfrac{\mathrm{pred}_i - \mathrm{ref}_i}{\mathrm{ref}_i}, & \text{minimization}, \\ \dfrac{\mathrm{ref}_i - \mathrm{pred}_i}{\mathrm{ref}_i}, & \text{maximization}. \end{cases} \tag{18}$$

We then map this gap to

$$\psi(\delta_i) = \begin{cases} \exp(-3\delta_i), & \delta_i \geq 0, \\ 1 + 0.5\tanh(-2\delta_i), & \delta_i < 0, \end{cases} \tag{19}$$

and add a declaration-match bonus $a_i \in \{0, 1\}$ when the model's stated objective agrees with the value recomputed from the parsed structure. The ungated objective reward is

$$\tilde{o}_i = 2\,(0.9\,\psi(\delta_i) + 0.1\,a_i)\,. \tag{20}$$

In the empirical study, we set $w_o = 1.0$ and $w_s = 0.3$ in $r_i = w_o o_i + w_s s_i$.

For the clipped objective used in training, we use a *length-normalized sequence surrogate ratio*. If T_i denotes the generated length,

$$\rho_i(\theta) = \exp\left(\frac{1}{T_i}\sum_{t=1}^{T_i}\left[\log \pi_\theta(y_{i,t} \mid x_i, y_{i,<t}) - \log \pi_{\text{old}}(y_{i,t} \mid x_i, y_{i,<t})\right]\right). \tag{21}$$

This quantity is the exponential of the average token-level log-likelihood ratio, equivalently the geometric mean of tokenwise likelihood ratios over the sampled sequence. It is therefore *not* the exact sequence importance ratio $\pi_\theta(y_i \mid x_i)/\pi_{\text{old}}(y_i \mid x_i)$; rather, it is a bounded-scale surrogate used for numerical stability across outputs of different lengths. CLASP and Dual-primal GRPO use this surrogate and disable additional reward scaling. The vanilla GRPO row in Table 1 follows the standard trainer objective, so it should be interpreted as a conventional system-level baseline; isolating the surrogate alone would require an additional sequence-surrogate vanilla GRPO ablation. Accordingly, the clipped objective below should be read as the practical training surrogate used in our constrained variants, not as a quantity inheriting the standard importance-sampling interpretation or PPO monotonic-improvement guarantees. The clipped surrogate term is

$$\ell_i(\theta) = -\min\left(\rho_i(\theta)A_i^{\text{CLASP}},\ \text{clip}(\rho_i(\theta), 1-\epsilon_{\text{low}}, 1+\epsilon_{\text{high}})\,A_i^{\text{CLASP}}\right), \tag{22}$$

optionally with a KL regularizer of weight κ_{KL}.

We use *Dual-primal GRPO* as the canonical name for the constrained GRPO baseline. It shares the same SFT initialization, prompt groups, dense cost components, length-normalized sequence ratio, and clipped surrogate family as CLASP, but it does not use the adaptive centered primal-gap grouped advantage in Eq. (6). Its sample advantage is

$$A_i^{\text{DP}} = \left(r_i - \bar{r}_G\right) - \sum_{j=1}^{m_\tau} \lambda_j^{(\tau)} h_{ij}^{(\tau)}, \tag{23}$$

with $h_{ij}^{(\tau)}$ denoting the signed masked gap introduced above; the clipped surrogate above is then evaluated with A_i^{DP} in place of A_i^{CLASP}. Thus the baseline centers only the primal reward within the prompt group and adds an uncentered controller-weighted signed-gap term, whereas CLASP centers both primal and controller-weighted positive-gap comparisons inside the adaptive grouped advantage.

Each task maintains its own controller $\boldsymbol{\lambda}^{(\tau)}$. It acts as a practical weighting mechanism for feasibility pressure rather than exact dual ascent for the population Lagrangian. Under the sign convention of Eq. (11), the update is a projected signed-gap controller. Because minibatches are mixed-task but controllers are task-specific, the controller statistics are computed separately for each task. Let $\mathcal{B}_t^{(\tau)}$ be the completions in the current controller window whose task is τ, and let $N_t^{(\tau)} = |\mathcal{B}_t^{(\tau)}|$. Every n_λ primal steps, the task-local masked batch statistics are aggregated as

$$\hat{h}_{t,j}^{(\tau)} = \frac{1}{N_t^{(\tau)}} \sum_{i \in \mathcal{B}_t^{(\tau)}} m_{ij}^{(\tau)}\left(c_{ij}^{(\tau)} - \epsilon_j^{(\tau)}\right), \tag{24}$$

$$\bar{f}_t^{(\tau)} = \frac{1}{N_t^{(\tau)}} \sum_{i \in \mathcal{B}_t^{(\tau)}} \mathbf{1}[v_i = 0]. \tag{25}$$

The controller then uses the feasibility-conditional step size

$$\eta_\lambda^{(t,\tau)} = \eta_0 \sqrt{\frac{1 - \bar{f}_t^{(\tau)} + \varepsilon}{\bar{f}_t^{(\tau)} + \varepsilon}} \tag{26}$$

Table 5: Evaluation questions and corresponding measurements. The table maps each empirical claim to the evidence reported in this manuscript, rather than to a complete possible ablation suite.

Claim	Evidence reported in this paper
CLASP-GRPO improves feasibility	Table 1 reports feasibility and conditional reference gaps for SFT+BoN, vanilla GRPO+BoN, Dual-primal GRPO+BoN, CLASP+BoN, and the Jiang et al. reference rows; Tables 3, 4, 11, and 12 add solver, scale-stratified, aggregate, and runtime context.
CLASP-GRPO improves best feasible quality	Table 1 reports mean best-feasible reference gaps with dispersion for BoN=8, separately from feasibility; Table 4 reports conditional gap and Success@$1/5/10$ by size bin; Tables 9 and 10 provide targeted Taillard JSSP and TSPLIB comparisons, including the reported TSPLIB BoN=8/64 distinction.
The proposed mechanism explains the gains	Mechanism evidence consists of the main hard-group diagnostic in Figure 1, the training-variant comparison in Appendix Figure 4, the SFT-vs-CLASP violation distributions in Figure 2, the appendix all-infeasible-group diagnostics in Figures 8 and 6, the mechanism dashboard in Figure 9, and the training-dynamics curves in Figure 10. Tables 6, 7, and 8, together with Appendix F, define the data, sequence-pressure diagnostics, training settings, adaptive length mechanism, dense costs, infeasible-sample credit, feasibility rule, and controller parameters used by those diagnostics.

and updates

$$\lambda_{t+1,j}^{(\tau)} = \Pi_{[0,\lambda_{\max}]}\left(\lambda_{t,j}^{(\tau)} + \eta_{\lambda}^{(t,\tau)}\hat{h}_{t,j}^{(\tau)}\right). \tag{27}$$

This rule reacts to the *current* policy rather than to an exponential moving average of past violations. The projection keeps the controller in a compact domain and the shared step size couples controller aggressiveness to the observed feasibility regime. As in the main text, this appendix-level description is intentionally algorithmic: it specifies the mechanism used in practice, not a general convergence guarantee for the overall non-convex training dynamics. For zero-tolerance components, the clipped costs have no negative interior margin and the corresponding controller weight can only increase or remain fixed, whereas positive-tolerance components can yield negative signed gaps and therefore allow the projected controller weight to decay.

CLASP also maintains a task-adaptive completion budget. For each task τ, a rolling window of recent decoded lengths is updated online; every R optimizer steps, the generation cap is reset to the clipped empirical quantile in Eq. (7). This leaves the reward definition unchanged, but it changes the effective compute allocation across tasks during RL.

C Claim Map and Experimental Setup

D Benchmark and Configuration Details

The primary training configuration uses a Qwen2.5-7B-Instruct base model with multi-node H100 GPU training, bfloat16 arithmetic, and LoRA adaptation. We keep the SFT stage fixed across methods and vary only the RL stage. In the reported five-task setup, the SFT stage uses 50,000 supervised examples per task, compared with 500,000 in the reproduced Jiang et al. reference setting. Table 8 reports the SFT initialization and the RL-stage hyperparameters that control the CLASP grouped advantage, controller dynamics, and adaptive length mechanism.

Some implementation flags and historical ablation logs retain early names such as `clasp_pareto_tau` and `clasp_pareto_anchor` for checkpoint and script compatibility. In the notation of this paper, `clasp_pareto_tau` is the transition-sharpness parameter β, and disabling the historical `clasp_pareto_anchor` flag disables the adaptive grouped advantage in Eq. (6); no Pareto anchor enters the mathematical grouped advantage.

Table 6: Benchmark suite used in this paper. The table summarizes the tasks, data volumes, and output schemas used in the supported experimental setup. Reported SFT train-set sizes correspond to the subsets actually used in the experiments, before within-split deduplication and train/validation partitioning; each task uses 50,000 supervised examples. The five-task setting trains and evaluates on TSP, OP, CVRP, MIS, and JSSP; MVC and PFSP appear only in legacy statistics files and are excluded from the experiments.

Task	Objective direction	SFT train	RL train	Eval	Output schema
TSP	minimize	50,000	3,200	300	`Route`, `Objective`
OP	maximize	50,000	5,000	300	`Route`, `Objective`
CVRP	minimize	50,000	2,000	300	`Routes`, `Objective`
MIS	maximize	50,000	3,200	300	`Set`, `Objective`
JSSP	minimize	50,000	3,200	300	`Schedule`, `Makespan`

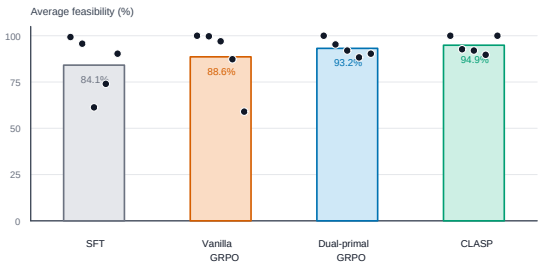


Figure 4: Average feasibility across the main training variants. Higher is better. The plotted comparison includes SFT, vanilla GRPO, Dual-primal GRPO, and CLASP; it should be read together with the main hard-group diagnostic in Figure 1, rather than as a complete factorial ablation suite.

Evaluation uses the same prompt template, parsers, and thresholded feasibility checks as training-time reward computation, but every candidate is rescored from its parsed discrete structure. For best-of-N decoding, we select the feasible candidate with the lowest recomputed reference gap for each instance; ties are broken by lower violation and then by the earliest sampled candidate. Instances with no feasible candidate contribute to feasibility and Success@k rates but are excluded from mean and median best-feasible gap summaries. We avoid assigning infeasible instances an arbitrary penalized optimality gap; when feasibility differs sharply, Success@k and feasibility are the fair all-instance comparisons, while best-feasible gap is a conditional quality diagnostic.

Table 7: Dataset and context diagnostics for the supported five-task setup. The table highlights sequence-length pressure and split overlap for the tasks used in the experiments. The token statistics aggregate only TSP, OP, CVRP, MIS, and JSSP, and the overlap count is based on a direct audit of the provided data splits.

Diagnostic	Value
Supported five-task combined sequences $>$ 4096 tokens	21.10%
Supported five-task combined sequences $>$ 8192 tokens	0.88%
Supported five-task combined sequences $>$ 12288 tokens	0.04%
TSP SFT-eval / GRPO-eval exact overlap	300
OP/CVRP/MIS/JSSP SFT-eval / GRPO-eval overlap	0

Table 8: Historical training hyperparameters retained for audit. The current reproduced comparison is defined by the source-of-truth manifest and the frozen launch receipts, not by this legacy table.

Category	Value
SFT initialization	Qwen2.5-7B-Instruct; bfloat16 arithmetic; no-reason Alpaca prompt style; seed 42; 50,000 supervised examples per task with evaluation ratio 0.05; LoRA rank 64, LoRA $\alpha = 64$, LoRA dropout 0; learning rate 2×10^{-4}; 5 epochs; context window / maximum sequence length 8192; per-device train/eval batch $2/1$, effective optimization batch 64 on 32 H100 GPUs.
RL model and adaptation	Initialized from the fixed SFT checkpoint; bfloat16 arithmetic; no-reason Alpaca prompt style; seed 42; full finetuning disabled; LoRA rank 64, LoRA $\alpha = 64$, LoRA dropout 0; learning rate 10^{-6}; 20 epochs.
Rollouts and surrogate	Rollout group size $K = 8$; per-device train/eval batch $4/4$; gradient accumulation 4, effective optimization batch 256 on 16 H100 GPUs; generation batch 512; maximum prompt length 7168; maximum completion length $L_{\max} = 1024$; context window 8192; constrained variants use the length-normalized sequence surrogate from Eq. (21) and PPO-style clipping as in Eq. (22), with no additional reward scaling.
Trainer defaults	Other trainer hyperparameters not overridden by the launchers, including optimizer type, scheduler, Adam coefficients, weight decay, gradient clipping, and PPO clipping constants, follow the TRL/Transformers defaults for the recorded TRL 0.24.0, Transformers 4.57.3, and Accelerate 1.12.0 environment.
CLASP grouped advantage and reward gate	Feasibility transition sharpness $\beta = 4.0$; within-group standard-deviation stabilizer $\varepsilon = 10^{-4}$; primal weight $w_o = 1.0$; structure weight $w_s = 0.3$; infeasible partial-credit gate from Eq. (3) enabled.
Controller	Base controller step size $\eta_0 = 0.0075$; controller refresh period $n_\lambda = 4$ primal optimizer steps; $\lambda_0 = 0$; $\lambda_{\max} = 1.0$; feasibility-conditioned step-size stabilizer $\varepsilon = 10^{-3}$.
Dense-cost thresholds	Global controller cost limit 0.03; violation tolerance 0.01; task-component tolerances are listed in Appendix F.
Adaptive length	Enabled; rolling-window size 1000; quantile $q = 0.99$; floor $L_{\min} = 128$; refresh period $R = 50$ optimizer steps.

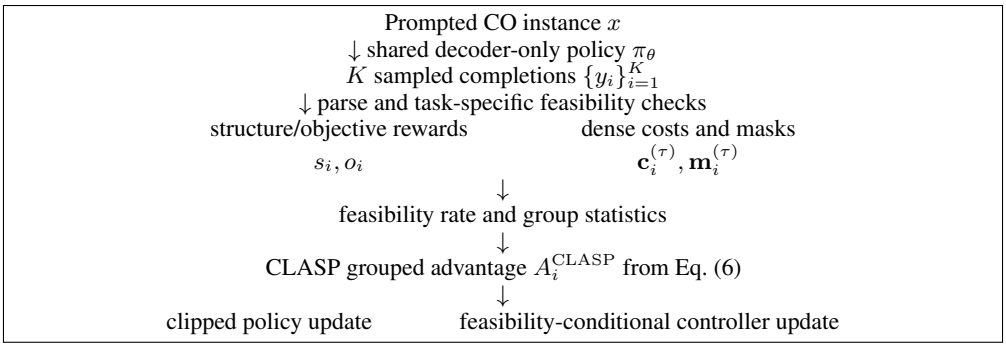


Figure 5: Overview of CLASP-GRPO, highlighting the adaptive centered within-group advantage, taskwise signed-gap controller, and adaptive decoding budget.

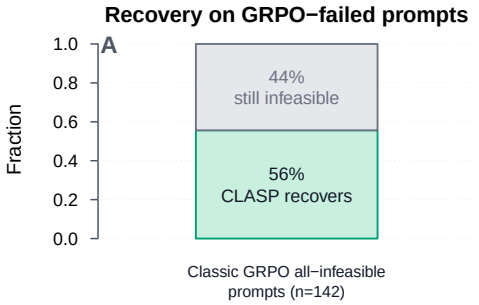


Figure 6: Recovery on prompts where classic GRPO produced no feasible candidate. The denominator is the 142 hard-split prompts for which all $K = 8$ classic GRPO candidates were infeasible; CLASP recovers at least one feasible candidate on 55.6% of them.

E Additional Reference Results and Diagnostics

Figure 3 is the authoritative current-result plot for this revision. The remaining PDF figures in this appendix are legacy diagnostics retained for auditability; their raw generation inputs were not present in the archived paper package and they must not be interpreted as plots of the current pending evaluation campaign. Figure 5 provides a conceptual visual summary of the CLASP-GRPO training loop.

F Task-Specific Dense Cost Components

The dense cost vectors used in this work are:

- **TSP**: `parse_failure`, `depot_boundary`, `invalid_node`, `duplicate_customer`, `missing_customer`, `interior_depot`.
- **CVRP**: `parse_failure`, `route_count_mismatch`, `depot_boundary`, `invalid_node`, `duplicate_customer`, `missing_customer`, `interior_depot`, `capacity_overflow`.
- **OP**: `parse_failure`, `start_at_depot`, `invalid_node`, `duplicate_visit`, `length_overflow`.
- **MIS**: `parse_failure`, `invalid_node`, `duplicate_node`, `edge_conflict`.
- **JSSP**: `parse_failure`, `machine_row_mismatch`, `row_length_mismatch`, `duplicate_job`, `missing_job`, `schedule_invalid`.

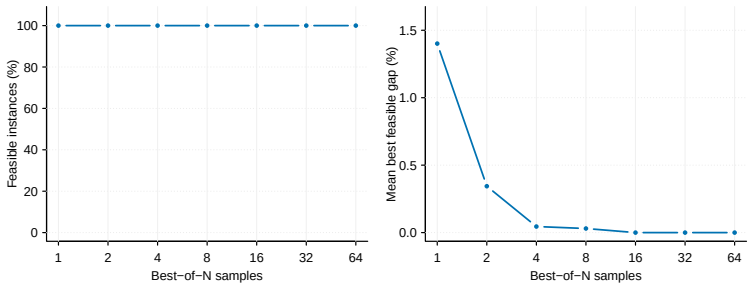


Figure 7: JSSP best-of-N ablation with varying candidate budget N. The plot reports how feasibility and conditional reference-value gap change as the best-of-N decoding budget increases from 1 to 64, providing appendix-level context for the BoN sensitivity of the JSSP results.

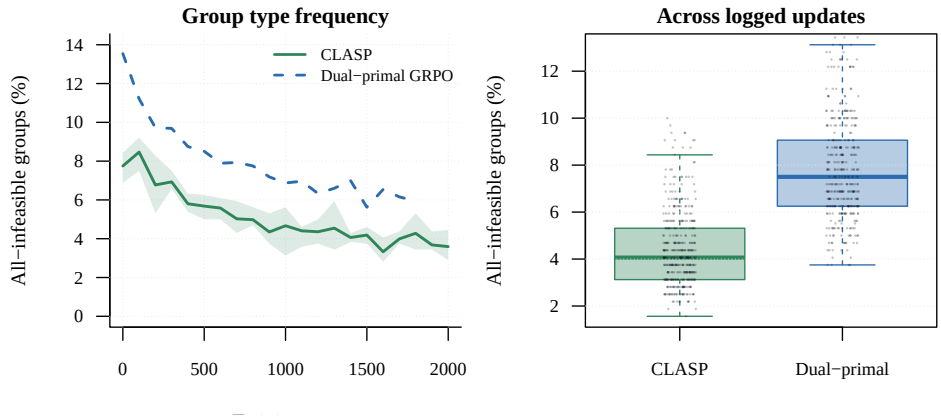


Figure 8: Frequency of hard prompt groups during training. Left: the fraction of all-infeasible groups over RL steps for CLASP and the Dual-primal GRPO baseline. Right: summary distribution of the all-infeasible-group rates across logged updates. The figure supports the motivating claim that hard prompts frequently induce all-infeasible groups and shows how the learned policies differ in that regime.

Every cost is clipped into $[0, 1]$, every mask is binary, and every sample must expose at least one active component.

- Tolerances use $\epsilon_j = 0$ for `parse_failure`, `depot_boundary`, `start_at_depot`, `invalid_node`, `duplicate_customer`, `missing_customer`, `duplicate_visit`, `duplicate_node`, `edge_conflict`, `machine_row_mismatch`, `row_length_mismatch`, `duplicate_job`, `missing_job`, `schedule_invalid`, and `interior_depot`; $\epsilon_j = 0.02$ for `route_count_mismatch` and `capacity_overflow`; and $\epsilon_j = 0.015$ for `length_overflow`.
- The binary feasibility flag is set to one only when the maximum positive component gap is at most 10^{-8}.
- The `co_default` weights are 5.0 for `parse_failure`, `missing_customer`, and `edge_conflict`; 4.0 for `invalid_node`; 3.5 for `depot_boundary`, `start_at_depot`, `machine_row_mismatch`, `row_length_mismatch`, and `duplicate_job`; 2.5 for `interior_depot`; 1.5 for `route_count_mismatch` and `duplicate_node`; 3.0 for `capacity_overflow` and `length_overflow`; 2.0 for `duplicate_visit`; and 4.5 for `missing_job`.
- Task-specific overrides use 6.0 for CVRP `missing_customer`, MIS `edge_conflict`, and JSSP `schedule_invalid`; 5.0 for CVRP `duplicate_customer` and JSSP `missing_job`; 4.0 for CVRP `capacity_overflow`, CVRP `depot_boundary`, OP `length_overflow`, and OP `start_at_depot`; and 1.0 for MIS `duplicate_node`.

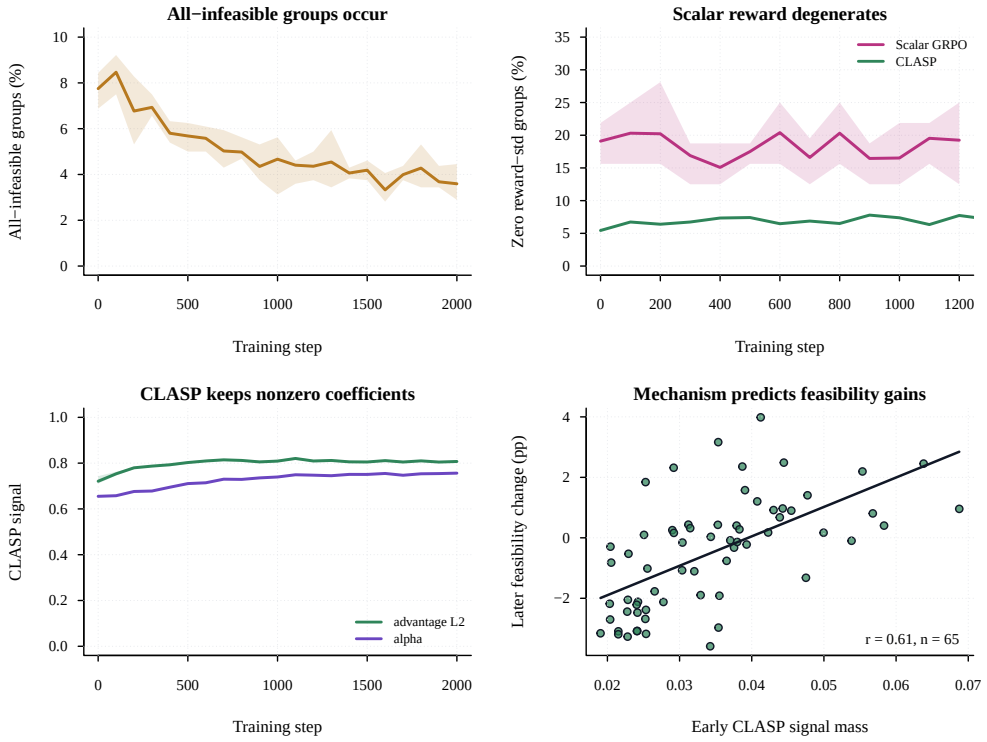


Figure 9: Mechanism diagnostics for CLASP. The panels summarize group-level failure modes and controller behavior, including all-infeasible-group prevalence, scalar-reward degeneracy, grouped-advantage evolution, and the relation between early CLASP gap reduction and later feasible-quality gains. Together they provide appendix-level evidence for the mechanism discussion in Section 3.

Under the mask-completeness assumption, the controller therefore always observes some task-defined violation information, even for malformed outputs.

For monitoring only, we also log the scalar diagnostic summary

$$\text{constraint}_i = 1 - \frac{1}{|\{j : m_{ij}^{(\tau)} = 1\}|} \sum_{j : m_{ij}^{(\tau)} = 1} c_{ij}^{(\tau)}, \tag{28}$$

which is well-defined because every sample exposes at least one active component. This diagnostic is not used in either the CLASP grouped advantage or the controller update.

G Formal Arguments and Proofs

Proposition 1 (Population constraints align with thresholded feasibility)**.** *Fix a task τ and assume the mask family is complete for the task semantics. For each component j, define*

$$V_{\tau,j}(\theta) = \mathbb{E}_{x \sim \mathcal{D}_\tau,\, y \sim \pi_\theta(\cdot|x)} \left[m_j^{(\tau)}(x,y) \max\left\{0, c_j^{(\tau)}(x,y) - \epsilon_j^{(\tau)}\right\}\right]. \tag{29}$$

Then $V_{\tau,j}(\theta) \geq 0$ for every j, and the conjunction $V_{\tau,j}(\theta) = 0$ for all active task components is equivalent to almost-sure thresholded feasibility of the sampled completions with respect to the dense costs and tolerances, i.e.,

$$\sum_{j=1}^{m_\tau} m_j^{(\tau)}(x,y) \max\left\{0, c_j^{(\tau)}(x,y) - \epsilon_j^{(\tau)}\right\} = 0 \quad \textit{for } \mathcal{D}_\tau \otimes \pi_\theta\textit{-almost every } (x,y). \tag{30}$$

Under mask completeness, this almost-sure condition coincides with thresholded feasibility for all semantically applicable violation types.



Figure 10: Training-dynamics comparison between CLASP and Dual-primal GRPO. The panels track relative trends in feasibility, feasible quality, controller values, and related diagnostics over optimization. This figure is included in the appendix because it is useful for attribution and reviewer inspection, but too dense for the main paper.

Proposition 2 (Centered CLASP grouped advantages are well-defined)**.** *For every finite prompt group $G = \{(r_i, g_i(\boldsymbol{\lambda}), v_i)\}_{i=1}^K$ with $K \geq 1$, finite r_i and $g_i(\boldsymbol{\lambda})$, well-defined feasibility indicators $\mathbf{1}[v_i = 0]$, $\beta \in (0, \infty)$, and $\varepsilon > 0$, the CLASP grouped advantages in Eq.* (6) *are finite and satisfy $\sum_{i=1}^K A_i^{\mathrm{CLASP}} = 0$.*

Remark 1 (Reference-sample cancellation). For any group-level reference index $a \in [K]$, including one selected from a Pareto frontier or any other deterministic rule, define the raw reference advantage

$$\widetilde{A}_i(a) = \alpha_G \frac{r_i - r_a}{\sigma_r^G + \varepsilon} + (1 - \alpha_G) \frac{g_a(\boldsymbol{\lambda}) - g_i(\boldsymbol{\lambda})}{\sigma_g^G + \varepsilon}. \tag{31}$$

Its within-group centered form is independent of a:

$$\widetilde{A}_i(a) - \frac{1}{K} \sum_{k=1}^K \widetilde{A}_k(a) = \alpha_G \frac{r_i - \bar{r}_G}{\sigma_r^G + \varepsilon} + (1 - \alpha_G) \frac{\bar{g}_G - g_i(\boldsymbol{\lambda})}{\sigma_g^G + \varepsilon} = A_i^{\mathrm{CLASP}}. \tag{32}$$

Thus no reference sample enters the mathematical grouped advantage in Eq. (6); historical implementation names involving "anchor" refer to enabling or disabling the whole centered grouped advantage.

Proposition 3 (Projected CLASP controller fixed-point conditions)**.** *Fix a task τ and freeze θ. Consider the deterministic controller obtained by replacing minibatch statistics with their expectations, and assume the signed moment $G_{\tau,j}(\theta)$ is finite:*

$$F_\tau(\theta) = \mathbb{E}_{x \sim \mathcal{D}_\tau,\, y \sim \pi_\theta(\cdot|x)} \big[\mathbf{1}[v(x, y) = 0]\big], \tag{33}$$

$$\lambda_{t+1,j} = \Pi_{[0,\lambda_{\max}]} \left(\lambda_{t,j} + \eta_0 \sqrt{\frac{1 - F_\tau(\theta) + \varepsilon}{F_\tau(\theta) + \varepsilon}}\, G_{\tau,j}(\theta) \right), \tag{34}$$

where $\eta_0 > 0$, $\varepsilon > 0$, $\lambda_{\max} > 0$, and $\Pi_{[0,\lambda_{\max}]}$ denotes Euclidean projection onto $[0, \lambda_{\max}]$. Any fixed point $\lambda_j^\star$ satisfies the sign conditions

$$0 < \lambda_j^\star < \lambda_{\max} \implies G_{\tau,j}(\theta) = 0, \tag{35}$$

$$\lambda_j^\star = 0 \implies G_{\tau,j}(\theta) \leq 0, \tag{36}$$

$$\lambda_j^\star = \lambda_{\max} \implies G_{\tau,j}(\theta) \geq 0. \tag{37}$$

Theorem 1 (Non-degeneracy of the CLASP grouped advantage in the constant-primal all-infeasible regime)**.** *Let $G = \{(r_i, g_i(\boldsymbol{\lambda}), v_i)\}_{i=1}^K$ be a finite prompt group with $K \geq 2$ and finite real-valued primal scores and controller-weighted gaps. Assume that:*

1. *every completion in the group is infeasible, so $v_i > 0$ for all $i \in [K]$;*
2. *the primal scores are constant across the group, so $r_i = r_\circ$ for all $i \in [K]$;*
3. *the controller-weighted feasibility gaps are not constant across the group, so there exist $i, j \in [K]$ with $g_i(\boldsymbol{\lambda}) \neq g_j(\boldsymbol{\lambda})$;*
4. *The dense-rank and recovery estimators use finite positive stabilizers and bounded rank scores, so their centered group credits are finite.*

Then the centered grouped advantages $(A_i^{\mathrm{CLASP}})_{i=1}^K$ defined in Eq. (6) *are well-defined, sum to zero, and are not identically zero. By contrast, any mean-centered scalar-reward advantage based only on $(r_i)_{i=1}^K$ and proportional to $r_i - \bar{r}_G$, including the stabilized standardized quantity $(r_i - \bar{r}_G)/(\sigma_r^G + \varepsilon)$, is identically zero on the same group.*

Proof of Theorem 1. By Proposition 2, the CLASP grouped advantages are finite and sum to zero. Because every completion is infeasible, $f_G = 0$, so $\alpha_G = \sigma(-\beta/2) \in (0, 1)$ for every finite $\beta > 0$. Because the primal scores are constant across the group, $\bar{r}_G = r_\circ$, $\sigma_r^G = 0$, and the primal contribution to Eq. (6) is identically zero. Because the controller-weighted gaps are not constant, $\sigma_g^G > 0$.

Therefore the CLASP grouped advantages reduce to

$$A_i^{\mathrm{CLASP}} = (1 - \alpha_G) \frac{\bar{g}_G - g_i(\boldsymbol{\lambda})}{\sigma_g^G + \varepsilon}. \tag{38}$$

Since $(g_i(\boldsymbol{\lambda}))_{i=1}^K$ is not constant, at least one $g_i(\boldsymbol{\lambda})$ differs from $\bar{g}_G$. For that index, $A_i^{\mathrm{CLASP}} \neq 0$ because $1 - \alpha_G > 0$ and $\sigma_g^G + \varepsilon > 0$. By contrast, the primal scores are constant, so $r_i - \bar{r}_G = 0$ for every i. Hence every mean-centered scalar-reward advantage proportional to $r_i - \bar{r}_G$, including $(r_i - \bar{r}_G)/(\sigma_r^G + \varepsilon)$, is identically zero. Therefore A^{CLASP} is well-defined, zero-sum, and non-trivial, whereas the centered scalar baseline is degenerate on the same group. □

Proof of Proposition 1. Each integrand in $V_{\tau,j}(\theta)$ is nonnegative, so $V_{\tau,j}(\theta) \geq 0$. If $V_{\tau,j}(\theta) = 0$, then the nonnegative random variable

$$Z_{\tau,j}(x,y) = m_j^{(\tau)}(x,y) \max\left\{0, c_j^{(\tau)}(x,y) - \epsilon_j^{(\tau)}\right\} \tag{39}$$

has zero expectation and therefore vanishes for $\mathcal{D}_\tau \otimes \pi_\theta$-almost every (x,y). If this holds for every active component j, then the finite sum of these nonnegative terms also vanishes almost surely, which is exactly the displayed thresholded-feasibility condition. Conversely, if the summed thresholded-feasibility condition holds almost surely, then each nonnegative summand must vanish almost surely, implying $V_{\tau,j}(\theta) = 0$ for every j. Under mask completeness, every semantically applicable violation type is represented by an active mask entry and corresponding dense cost, so this almost-sure vanishing condition coincides with thresholded feasibility for all semantically applicable violation types. □

Proof of Proposition 2. The logistic sigmoid maps the finite argument $\beta(f_G - 1/2)$ to a finite $\alpha_G \in (0,1)$. The denominators $\sigma_r^G + \varepsilon$ and $\sigma_g^G + \varepsilon$ are strictly positive because empirical standard deviations are nonnegative and $\varepsilon > 0$. Thus each grouped advantage in Eq. (6) is finite whenever the group scores are finite. Moreover,

$$\sum_{i=1}^K (r_i - \bar{r}_G) = 0, \qquad \sum_{i=1}^K (\bar{g}_G - g_i(\boldsymbol{\lambda})) = 0. \tag{40}$$

Multiplying these two zero sums by the group-constant factors in Eq. (6) gives $\sum_{i=1}^K A_i^{\mathrm{CLASP}} = 0$. □

Proof of Proposition 3. Fix a component j. The scalar factor

$$\kappa_\tau(\theta) = \eta_0 \sqrt{\frac{1 - F_\tau(\theta) + \varepsilon}{F_\tau(\theta) + \varepsilon}} \tag{41}$$

is finite and strictly positive: $F_\tau(\theta) \in [0,1]$ because it is an expectation of an indicator, while $\eta_0 > 0$ and $\varepsilon > 0$. At a fixed point,

$$\lambda_j^\star = \Pi_{[0,\lambda_{\max}]}\left(\lambda_j^\star + \kappa_\tau(\theta) G_{\tau,j}(\theta)\right). \tag{42}$$

If $0 < \lambda_j^\star < \lambda_{\max}$, the projection is inactive at $\lambda_j^\star$, so Eq. (42) implies $\kappa_\tau(\theta) G_{\tau,j}(\theta) = 0$. Since $\kappa_\tau(\theta) > 0$, we obtain $G_{\tau,j}(\theta) = 0$.

If $\lambda_j^\star = 0$, Eq. (42) can hold only if the quantity inside the projection is non-positive. That is, $\kappa_\tau(\theta) G_{\tau,j}(\theta) \leq 0$. Positivity of $\kappa_\tau(\theta)$ then gives $G_{\tau,j}(\theta) \leq 0$.

If $\lambda_j^\star = \lambda_{\max}$, the quantity inside the projection must be at least $\lambda_{\max}$, which implies $\kappa_\tau(\theta) G_{\tau,j}(\theta) \geq 0$. Again $\kappa_\tau(\theta) > 0$, so $G_{\tau,j}(\theta) \geq 0$. This proves the stated complementary-sign conditions. □

H Appendix Discussion and Supplemental Tables

Tables 9, 10, 11, and 12 provide a compact appendix-level view of how the proposed method compares under four complementary lenses. Table 9 shows that the matched Taillard-format JSSP comparison remains competitive but still sensitive to evaluation protocol and candidate budget, so we interpret it as a targeted external reference rather than a standalone headline result. Figure 7 complements that table with a JSSP BoN ablation study over varying N, highlighting how larger decoding budgets affect feasibility and gap. Table 10 adds a TSPLIB TSP comparison against prior LLM-based methods and shows that increasing CLASP's best-of-N budget substantially narrows the gap.

Tables 11 and 12 summarize the broader positioning of CLASP across problems and runtime. Together they show that CLASP remains a fast learned policy and matches or improves the Jiang et al. best-of-N reference gaps across the five shared tasks under the current protocol. They also show that CLASP preserves the fast policy-decoding profile of the underlying SFT/RL model, even when it is modestly slower than the lightest open-weight general-purpose models on some tasks.

Table 9: Reference comparison on Taillard-format JSSP instances from Jiang et al. [2025]. Lower objective value and lower relative gap indicate better performance. Rows labeled "Jiang et al." correspond to entries labeled "Ours" in the source paper, and asterisks follow the source notation. Comparisons should be interpreted only under matched N and evaluation protocol."Adj.-swap polish" denotes a deterministic post-processing step that starts from the best feasible BoN schedule and greedily applies adjacent swaps in machine-order lists whenever the swap reduces the JSSP makespan.

Method	TA 15x15		TA 20x15		TA 20x20	
	obj.↓	opt.↓	obj.↓	opt.↓	obj.↓	opt.↓
Best-known solution	1228.9	–	1364.9	–	1617.3	–
SPT	1546.1	25.81%	1813.5	32.87%	2067.2	27.81%
FIFO	1657.4	34.87%	2008.4	47.15%	2297.1	42.03%
ATC	1586.8	29.12%	1794.3	31.46%	2114.1	30.72%
GA* 100x100	1583.8	28.88%	1847.1	35.23%	2304.2	42.47%
L2D greedy	1522.3	23.88%	1813.4	32.86%	2130.2	31.21%
L2D sample	1543.8	25.62%	1779.7	30.39%	2134.2	31.96%
StarJob*	–	19.68%	–	26.91%	–	33.12%
Jiang et al. (N = 8)	1403.6	14.22%	1653.7	21.16%	1994.2	23.30%
Jiang et al. (N = 64)	1368.2	11.34%	1600.5	17.26%	1940.9	20.01%
CLASP (N = 8)	1410.6	14.79%	1666.2	22.00%	2038.0	25.80%
CLASP (N = 64)	1402.9	14.16%	1643.4	20.32%	2009.8	24.06%
CLASP + adj.-swap polish ($N = 64$)	**1357.7**	**10.48%**	**1554.0**	**13.85%**	**1907.1**	**17.92%**

Table 10: Optimality gap comparison of LLM-based methods on TSPLIB instances. CLASP uses normalized-coordinate TSPLIB prompts; reported CLASP gaps are recomputed on the original TSPLIB coordinates with `EUC_2D` rounded distances.

Instance	AEL	ReEVO	MCTS-AHD	Jiang et al.	CLASP (BoN=8)	CLASP (BoN=64)
eil51.tsp	10.84%	6.50%	15.98%	2.02%	3.29%	1.64%
berlin52.tsp	12.91%	17.99%	7.08%	4.54%	4.07%	0.08%
st70.tsp	4.10%	6.42%	15.45%	0.00%	2.67%	0.15%
eil76.tsp	10.26%	7.34%	12.46%	2.91%	2.60%	1.12%
pr76.tsp	9.06%	12.17%	10.41%	3.87%	3.91%	2.48%
Average Gap	9.43%	10.08%	12.28%	2.69%	3.31%	1.09%

Table 11: Performance comparison across different optimization problems.

Method	TSP	OP	CVRP	MIS	JSSP
SOTA method	0.00%	1.60%	-0.19%	0.00%	2.45%
Gurobi (1s)	5.76%	49.42%	-	2.28%	41.34%
Gurobi (10s)	1.82%	21.96%	35.09%	0.46%	2.45%
Jiang et al.	1.07%	1.85%	4.53%	1.04%	8.20%
CLASP	1.01%	1.18%	4.18%	1.04%	8.19%

Table 12: Detailed average wall-clock solving time per problem instance for the evaluation setting used in the corresponding accuracy tables. Rows with best-of-N decoding, including CLASP, include decoding, parsing, rescoring, and feasible-candidate selection.

Method	TSP	OP	CVRP	MIS	JSSP
General-purpose Language Models					
GPT-4o	7.5s	4.8s	6.8s	3.7s	5.8s
GPT-4o-mini	5.1s	2.4s	5.5s	3.2s	9.0s
Claude-3.7-Sonnet	6.7s	4.4s	6.7s	3.6s	9.2s
Claude-3.5-Haiku	5.0s	4.3s	3.5s	4.9s	9.0s
DeepSeek-V3-671B	33.0s	22.8s	36.3s	20.3s	26.2s
Llama3.3-70B	2.6s	1.8s	2.2s	1.8s	3.1s
Qwen2.5-72B	13.5s	13.3s	13.6s	6.9s	22.1s
Reasoning Models					
GPT-o3-mini	57.2s	109.8s	60.5s	109.4s	45.9s
GPT-o1	118.7s	120.7s	72.1s	293.5s	98.2s
DeepSeek-R1	240.0s	521.3s	360.5s	402.9s	364.8s
Prompt Strategies					
OPRO	150.2s	123.7s	121.6s	103.1s	191.2s
LMEA	418.8s	276.7s	415.2s	241.9s	432.0s
PHP	96.1s	95.9s	104.2s	85.3s	125.2s
SGE	166.9s	234.3s	308.8s	163.7s	260.5s
Jiang et al.					
SFT	7.0s	4.6s	6.9s	3.2s	10.7s
SFT+RL	7.0s	4.6s	6.9s	3.2s	10.7s
SFT+RL+BoN	11.7s	10.8s	11.3s	4.0s	20.7s
CLASP	7.34s	5.28s	7.96s	3.28s	11.72s

I Additional Performance Comparison

Tables 11 and 12 collect the cross-problem summary and the runtime comparison in one place for quick reference.

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